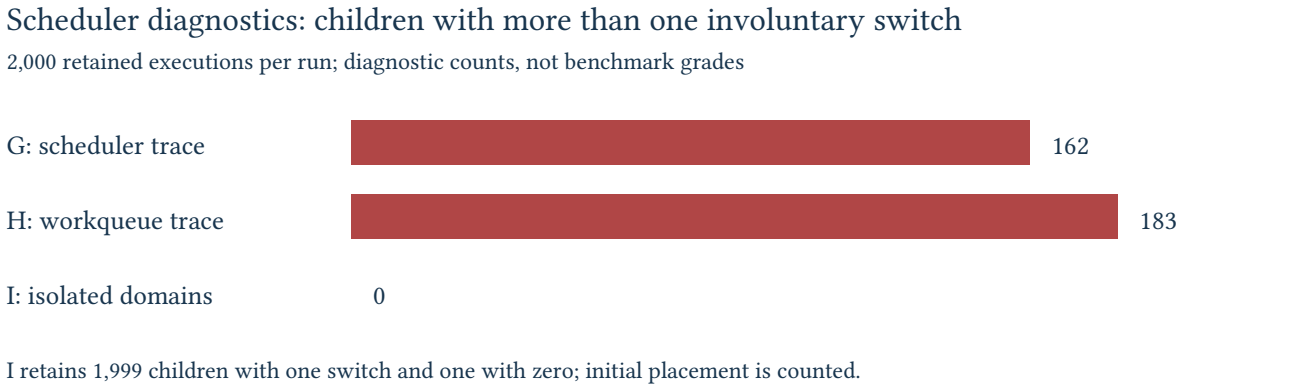
Dedicated benchmark workers: qualified L2 bootstrap

Inspected PDF (report.pdf) · native verification (verification.json) · deployment (deployment.json) · raw evidence (runs.tar.xz)

Qualification J passed the unchanged native gates: 3,606 verified L2 executions, 3,600 matching measured replays, 1,200 admitted A/A pairs and all 19 S3 payload hashes verified. The qualified bootstrap is deployed as profile v6 on the dedicated cpu-l2-spot pool. Ordinary cpu-spot retains its previous v4 bootstrap. The post-merge main-only GitHub Action is still pending final review and CI.



Diagnosis and retained attempts

Attempt	What actually ran	Outcome
F	No Job or trace: the completed E worker had auto-consolidated	SSM InvalidInstanceId; protocol and failure retained
G	2,000 native probe children, CPU4 scheduler trace	162 children had more than one involuntary switch; no excess busy ticks above two
H	2,000 children, scheduler/workqueue/timer trace	183 children had more than one switch; no excess busy ticks above two
I	2,000 children on a fresh candidate bootstrap, scheduler/workqueue trace	0 children above one switch; 0 traced native-probe preemptions on CPU4; diagnostic only
J	Full untraced crypto qualification on I's worker	All native L2, replay, A/A and S3 gates passed

Every G/H/I run retained all 2,000 samples and verified all six S3 payload hashes. G's 30,663-byte complete trace attributed native child preemptions to kworker/4:1 (150), migration/4 (11) and ksoftirqd/4 (1). H's 199,693-byte trace retained all 1,293 entries with no buffer loss. Its workqueue events included psi_avgs_work (183) and vmstat_update (61), plus less frequent filesystem/memory cleanup. These event counts are not one-to-one causal attribution for every switch. The initial RCU-only hypothesis did not explain the dominant observed work.

I's raw wait4 involuntary-switch histogram is zero: 1, one: 1,999. The probe pins after exec, so its counters include the initial affinity transition; ecbench pins in pre_exec. No zero-total-switch claim is made. I's complete five-entry trace contains no native-probe preemptions; CNI switches and a check_lifetime callback occur after the probe. Tracing overhead makes G/H/I diagnostic only. J began after explicit readback confirmed all diagnostic tracing disabled.

Qualified bootstrap and deployed contract

The candidate keeps periodic scheduler ticks, isolates worker scheduler domains and managed IRQs, offloads RCU callbacks, and binds RCU threads and watchdogs to housekeeping CPUs. cgroup_disable=pressure removes per-cgroup PSI work while mandatory host /proc/pressure/cpu and /proc/pressure/memory remain enabled. Every J record retains real before/after host pressure values. Existing strict whole-core static CPU management, NUMA, IRQ/workqueue and service placement remain. No native grader, threshold, correctness check or sample-admission rule changed.

Kernel sources explain why isolated domains avoid periodic vmstat shepherd work and why per-cgroup PSI can schedule delayed work. Sources: CPU isolation (https://docs.kernel.org/admin-guide/cpu-isolation.html), kernel threads (https://docs.kernel.org/admin-guide/kernel-per-CPU-kthreads.html), kernel parameters (https://www.kernel.org/doc/html/v6.13/admin-guide/kernel-parameters.html). The inspected Linux v6.18 source files are retained in the evidence archive.

Domain isolation changes ordinary unaffined multithreaded scheduling, so L2 uses its own tainted pool. submit-job --isolation-profile l2 selects cpu-l2-spot, profile v6 and its measurement toleration; standard jobs select only cpu-spot. Both pools allow broad x86 C/M/R Spot families, generation six or newer, subject to existing requirements and capacity. Each is capped at 32 vCPUs / 128 GiB: combined limits remain 64 vCPUs / 256 GiB under the regional 96-vCPU quota. Each node must fit its individual pool limit. Capacity is not guaranteed.

Deployment source: cc858fd93ab203f877c34b14205a0f22e554bf73. The production cpu-l2 class spec equals the qualified I/J spec; the ordinary cpu class equals the prior v4 spec. Both classes/pools were read back Ready with current generations as the isolab:submitters group. Both remote launchers were updated through CloudFormation and verified against immutable S3 versions and full archive hashes. Runtime checkpoint/archive pins did not change. The temporary diagnostic resources and this task's unchanged temporary cluster-admin association were removed; other tasks' jobs and workers were not paused, cancelled or resized.

Launcher	Immutable S3 package version
Ordinary remote launcher	GAbpTmZE8VYoN72rYIOwYqaior1Pj6ep
Recovery controller	FhkDu4kbpsBNUbxuDW16EYHRfTX3OQZe

Full native qualification J

Public 32-bit prime_search seed 59297; two targets with target seed 20261002; rho.negation reference, rho.plain baseline and rho.negation A/A control. Exactly 600 rounds, one warmup, alternating order, seed 1 and 120-second child timeout. Four exclusive CPUs, 4 GiB, 1,800 seconds including provisioning, fixed 60-second setup and zero retries. Boot readback completed during setup; no diagnostic exec ran during measurement. The worker was c8a.2xlarge, eight cores, one thread per core, AL2023 AMI ami-082b570a48503ea93, kernel 6.18.54, us-east-2c.

Gate	Native outcome
Correctness and isolation	3,606 / 3,606 verified and L2
Measured replay	3,600 / 3,600 matched
A/A	1,200 pairs admitted, zero below L2, outside_noise=false
A/A median ratio	1.0000835088461724
Native A/A 95% interval	[0.9998485970859599, 1.0002419602383978]
S3	19 / 19 full payload hashes matched

- Job: ecbench-domain-periodic-20261023-j / c97aa0ac-d3f6-431d-915d-189d21cd3c36.
- Pod UID: 76931218-b9ab-446c-a4b9-05dc068aa557.
- Worker: i-08df3df7752a77c99; node ip-10-84-74-123.us-east-2.compute.internal.
- Crypto source: cb2ec6873f30b683f7885b53c410d69965f02c20.
- Actual image: 104186811093.dkr.ecr.us-east-2.amazonaws.com/isolab/tasks@sha256:67eea58f5d4a7a8882b55f089c6063bb49fad58d40260ee9b0ceb349fc0e4581.
- Binary SHA-256: 583e62f0836eae43330c4b36bbe3d145dacb01de3132436c71f6924445c6a5a9.
- Results: s3://isolab-results-104186811093-us-east-2/runs/ecbench-domain-periodic-20261023-j/76931218-b9ab-446c-a4b9-05dc068aa557/.

Corrections, limits and reproduction

The frozen workload always used two targets and 600 rounds, producing 1,200 A/A pairs. The protocol prose and operational verifier initially expected 600 pairs; the original, exact-count correction and verifier are retained. No workload, threshold or sample changed. The first verifier invocation also retained an old D directory path and failed before reading evidence; that observer mistake is explicitly recorded. Native results were neither retried nor regraded.

G/H's initial SSM stdout hit its 24,000-byte limit. Complete traces were then retrieved as compressed data (H in two chunks) and checked against recorded compressed/plain hashes. Partial originals remain labeled and retained. The local MCP deployment initially hit ETXTBSY when replacing a running executable; atomic rename installed the verified binary without killing a running client.

This qualifies one fresh crypto workload/bootstrap combination. It does not prove all instance families, arbitrary CRIU portability, a new CRIU restore under v6, an algorithm speedup or completion of the main GitHub Action. Prior D provisioning (../20261023-l2-qualification-d/REPORT.md) and E preemption (../20261023-l2-qualification-e/REPORT.md) failures remain frozen. Earlier recovery evidence remains separate; interrupted timing is never admitted.

Run sha256sum -c runs.tar.xz.sha256 && tar -xJf runs.tar.xz. The archive retains all F/G/H/I/J protocols, raw native artifacts, full traces, manifests, instance and Job/Pod receipts, corrections, bootstrap, source observations and deployment receipts. Run sha256sum -c SHA256SUMS inside each run's artifacts directory to verify its complete payload set. Package binaries remain in immutable S3 objects identified by receipt; private checkpoint memory and credentials are not included. The graph derives counts from retained native rusage records and uses a common linear count scale; it computes no new timing statistic. Algorithm leaderboards and curve graphics were checked for scope and left unchanged. This evidence directory is frozen; later validation gets its own report.